

RED PIZZA





INTRODUCTION RED PIZZA

RedPizza is a growing pizza chain that aims to optimize its sales strategies by analyzing past order data. This SQL project focuses on key metrics to gain insights into order patterns, revenue generation, and customer preferences.



OBJECTIVE

The primary objective of this analysis is to extract meaningful insights from RedPizza's sales data using SQL queries. By understanding customer preferences and sales trends, the company can make informed decisions to boost revenue and improve customer satisfaction.

● ANALYSIS & INSIGHTS ●

1. TOTAL NUMBER OF ORDERS

```
SELECT  
    COUNT(order_id) AS Total_Orders  
FROM  
    orders;
```

Result Grid	
	Total_Orders
	21350

A query was executed to retrieve the total number of orders placed.
This metric helps in understanding the overall demand.

● ANALYSIS & INSIGHTS ●

2.TOTAL REVENUE GENERATED

```
• SELECT  
  ROUND(SUM(order_details.quantity * pizzas.price),  
        2) AS Total_revenue  
FROM  
  order_details  
  JOIN  
  pizzas ON pizzas.pizza_id = order_details.pizza_id
```

Result Grid	
	Total_revenue
▶	817860.05

The sum of all pizza sales was calculated to determine the total revenue. This provides insight into business performance

● ANALYSIS & INSIGHTS ●

3. HIGHEST-PRICED PIZZA

```
SELECT  
    pizza_types.name, pizzas.price  
FROM  
    pizza_types  
    JOIN  
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id  
ORDER BY pizzas.price DESC  
LIMIT 1;
```

Result Grid			Filter Row
	name	price	
▶	The Greek Pizza	35.95	

The most expensive pizza on the menu was identified, which can be useful for premium marketing strategies.

● ANALYSIS & INSIGHTS ●

4.MOST COMMON PIZZA SIZE ORDERED

```
SELECT
  pizzas.size,
  COUNT(order_details.order_details_id) AS order_count
FROM
  pizzas
  JOIN
  order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC;
```

Result Grid				
	size	order_count		
▶	L	18526		
	M	15385		
	S	14137		
	XL	544		
	XXL	28		

By analyzing the frequency of pizza sizes ordered, RedPizza can adjust inventory and pricing strategies accordingly.

● ANALYSIS & INSIGHTS ●

5.TOP 5 MOST ORDERED PIZZA TYPES

```
SELECT
  pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
  pizza_types
  JOIN
  pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
  JOIN
  order_details ON order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name
order by quantity DESC
limit 5;
```

Result Grid   Filter		
	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

The most frequently ordered pizzas were identified along with their quantities, revealing customer preferences.

● ANALYSIS & INSIGHTS ●

6. TOTAL QUANTITY OF EACH PIZZA CATEGORY ORDERED

```
• SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid   Filter		
	category	quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

By joining necessary tables, the sales performance of different pizza categories was analyzed.

● ANALYSIS & INSIGHTS ●

7. ORDER DISTRIBUTION BY HOUR OF THE DAY

```
SELECT  
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count  
FROM  
    orders  
GROUP BY HOUR(order_time);
```

Result Grid			Filter
	hour	order_count	
▶	11	1231	
	12	2520	
	13	2455	
	14	1472	
	15	1468	
	16	1920	
	17	2336	
	18	2399	
	19	2009	
	20	1642	
	21	1198	
	22	663	
	23	28	
	10	8	
	9	1	

Understanding peak ordering hours helps in staffing and promotional planning.

● ANALYSIS & INSIGHTS ●

8.CATEGORY-WISE DISTRIBUTION OF PIZZAS

```
• SELECT  
    category, COUNT(name)  
FROM  
    pizza_types  
GROUP BY category
```

Result Grid			Filter Rows
	category	count(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

A breakdown of pizza categories sold helps in analyzing product performance.

● ANALYSIS & INSIGHTS ●

9. AVERAGE NUMBER OF PIZZAS ORDERED PER DAY

```
SELECT  
    ROUND(AVG(quantity), 0) AS avg_pizza_ordered_per_day  
FROM  
    (SELECT  
        orders.order_date, SUM(order_details.quantity) AS quantity  
    FROM  
        orders  
    JOIN order_details ON orders.order_id = order_details.order_id  
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid		Filter Rows:
	avg_pizza_ordered_per_day	
▶	138	

This metric provides insights into daily demand trends.

● ANALYSIS & INSIGHTS ●

10.TOP 3 MOST ORDERED PIZZA TYPES BASED ON REVENUE

```
SELECT
  pizza_types.name,
  SUM(order_details.quantity * pizzas.price) AS revenue
FROM
  pizza_types
  JOIN
  pizzas ON pizzas.pizza_type_id = pizza_types.pizza_type_id
  JOIN
  order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	

Identifying high-revenue pizzas helps in strategic marketing and promotions.

● ANALYSIS & INSIGHTS ●

11. PERCENTAGE CONTRIBUTION OF EACH PIZZA TYPE TO TOTAL REVENUE

```
SELECT
    pizza_types.category,
    CONCAT(ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
        ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS Total_revenue
    FROM
        order_details
        JOIN
        pizzas ON pizzas.pizza_id = order_details.pizza_id) * 100,
        '%') AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC;
```

Result Grid			Filter
	category	revenue	
▶	Classic	26.91%	
	Supreme	25.46%	
	Chicken	23.96%	
	Veggie	23.68%	

This helps in evaluating the impact of each pizza type on overall sales.

● ANALYSIS & INSIGHTS ●

12.CUMULATIVE REVENUE OVER TIME

```
select order_date,  
sum(revenue) over(order by order_date) as cum_revenue  
from  
(select orders.order_date,  
sum(order_details.quantity*pizzas.price) as revenue  
from order_details join pizzas  
on order_details.pizza_id=pizzas.pizza_id  
join orders  
on orders.order_id=order_details.order_id  
group by orders.order_date) as sales;
```

Result Grid			Filter Rows:
order_date	cum_revenue		
2015-01-01	2713.8500000000004		
2015-01-02	5445.75		
2015-01-03	8108.15		
2015-01-04	9863.6		
2015-01-05	11929.55		
2015-01-06	14358.5		
2015-01-07	16560.7		
2015-01-08	19399.05		
2015-01-09	21526.4		
2015-01-10	23990.350000000002		
2015-01-11	25862.65		
2015-01-12	27781.7		
2015-01-13	29831.300000000003		
2015-01-14	32358.700000000004		
2015-01-15	34343.500000000001		
2015-01-16	36937.650000000001		

Analyzing cumulative revenue trends assists in financial forecasting.

● ANALYSIS & INSIGHTS ●

13.TOP 3 MOST ORDERED PIZZA TYPES BY REVENUE IN EACH CATEGORY

```
select name, revenue
from
(select category, name, revenue,
rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity)*(pizzas.price)) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id=pizzas.pizza_type_id
join order_details
on order_details.pizza_id=pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <=3;
```

Result Grid  Filter Rows:		
	name	revenue
▶	The Thai Chicken Pizza	43434.25
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41409.5
	The Classic Deluxe Pizza	38180.5
	The Hawaiian Pizza	32273.25
	The Pepperoni Pizza	30161.75
	The Spicy Italian Pizza	34831.25
	The Italian Supreme Pizza	33476.75
	The Sicilian Pizza	30940.5
	The Four Cheese Pizza	32265.700000000065
	The Mexicana Pizza	26780.75
	The Five Cheese Pizza	26066.5

Identifying the best-selling pizzas within each category helps refine product offerings.



CONCLUSION

The SQL analysis of RedPizza's sales data provides valuable insights into ordering trends, customer preferences, and revenue distribution. These findings can help the company optimize inventory, enhance marketing efforts, and improve overall sales performance. Future recommendations include:

- Offering promotions on high-demand pizzas during peak hours.
- Adjusting inventory based on frequently ordered sizes and categories.
- Targeted marketing for premium-priced pizzas to boost revenue.

By leveraging data-driven decisions, RedPizza can continue its growth and maintain a competitive edge in the market.





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THANK YOU!

